



GROUP 4

DETECTION OF PERSONAL PROTECTIVE EQUIPMENT USING SEMI-SUPERVISED LEARNING WITH YOLOV9-C

-
1. BRYAN VALLERYAN ALVONSO (00000114797)
 2. AMANDA ALIYAH (00000135964)
 3. AISHA REISSANI SOPYAN (00000136172)
 4. KENNY VALENT WINALDA SEMBIRING (00000136177)
 5. VAUROLLA AZ ZAHRA (00000139045)



BACKGROUND

- Around 56% of construction accidents are caused by the failure to use Personal Protective Equipment (PPE).
- There were more than 67,000 occupational injury cases in 2016.
- Safety monitoring is still carried out manually, making it a complex task and difficult to perform consistently.
- Manual supervision has limitations due to human distraction and crowded work environments.

- Main challenges:
- High annotation cost due to the time and expense required for manual data labeling.
 - Black-box nature of the model (lack of explainability in AI decision-making).





RESEARCH GAP

Author	Judul	Metode	Gap
Nikunj, et al (2025)	Edge-Based AI Solution for Helmet Compliance Monitoring with YOLOv9	YOLOv9 deployed on edge devices	Only focuses on one type of PPE (helmet) and does not comprehensively evaluate compliance levels for multiple PPE types.
Xin li, et al (2025)	OAM-YOLO: A Real-Time Small Object Detection Framework for PPE Compliance Monitoring in Industrial Environments	YOLO with attention module and multi-scale feature extraction	Fully dependent on supervised learning, requiring large annotated datasets with no labeling efficiency approach.
Nur Amielia,et al (2024)	A Real-Time Detection of Personal Protective Equipment Using Deep Learning for Construction Safety Monitoring	YOLO-based deep learning detection	Only detects PPE objects without a systematic or rule-based worker compliance analysis mechanism.
Kumar et al. (2021)	YOLOv4 Algorithm for the Real-Time Detection of Fire and Personal Protective Equipments at Construction Sites	YOLOv4 and YOLOv4-tiny (Darknet framework)	Limited to 6 classes, relies on fully supervised learning with 14,500 labeled data, and lacks Explainable AI (XAI) validation.



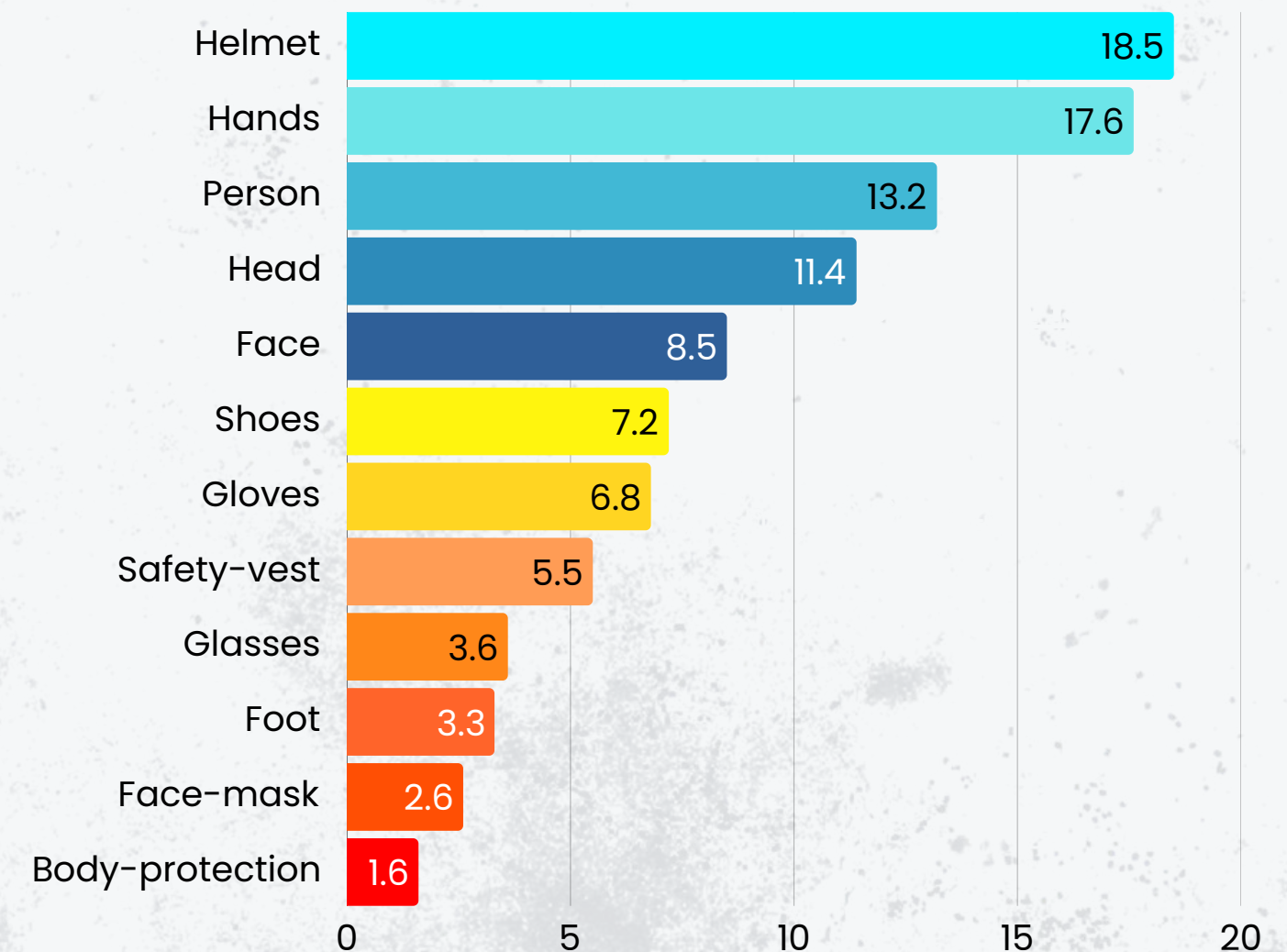
GROUP 4

DATASET

Combined from 8 public datasets.



Class Distribution



Total Images

22,033

Annotations

104,929

Classes

12

Train

18,956

Validation

4,775

Max Imbalance
Ratio

11,7x



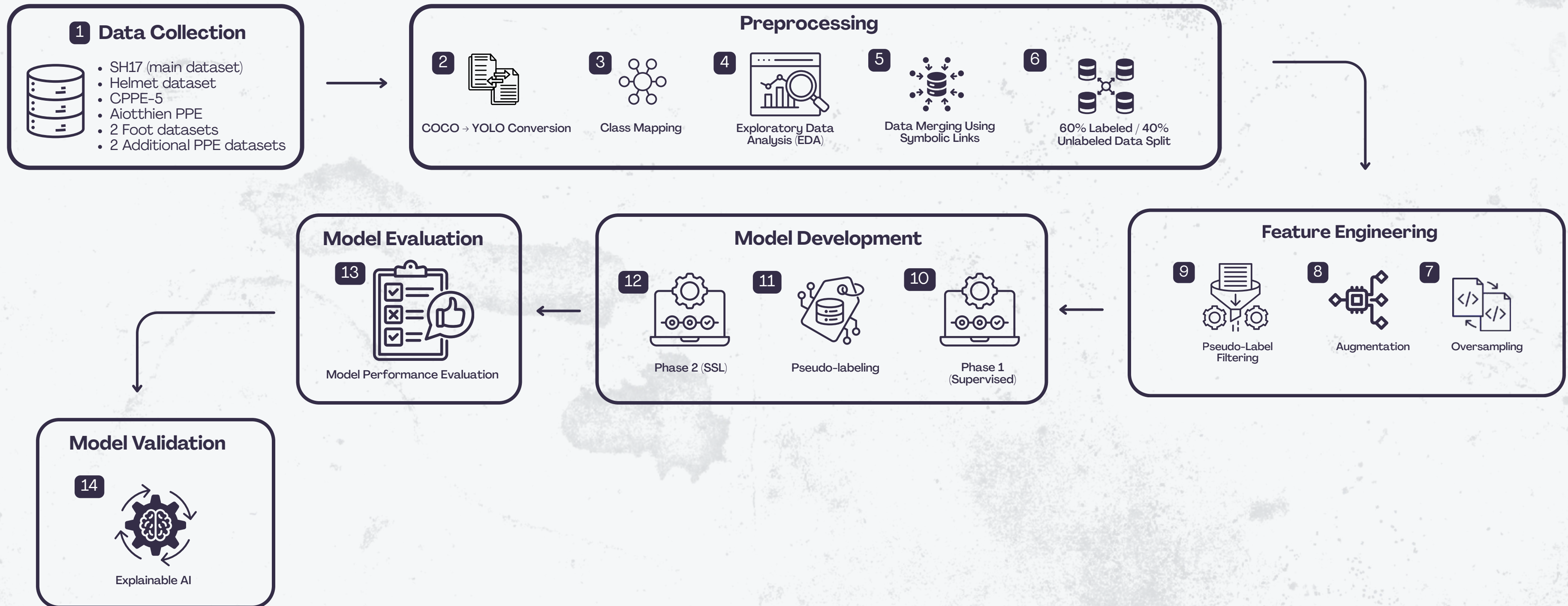
GROUP 4

RESEARCH PIPELINE



GROUP 4

RESEARCH PIPELINE





GROUP 4

PRE-PROCESSING + FEATURE ENGINEERING



DATA PRE-PROCESSING

1. COCO → YOLO Conversion

```
# Normalisasi ke format YOLO: <class> <cx> <cy> <bw> <bh>
cx = (x + w / 2) / img_info["width"]
cy = (y + h / 2) / img_info["height"]
bw = w / img_info["width"]
bh = h / img_info["height"]

new_cls = CLASS_MAPPING[cat]

# Tulis ke file .txt (satu file per gambar)
stem = Path(img_info["file_name"]).stem
lbl_file = CPPE_OUTPUT / "labels" / f"{stem}.txt"

with open(lbl_file, "a") as f:
    f.write(f"{new_cls} {cx:.6f} {cy:.6f} {bw:.6f} {bh:.6f}\n")
```

CPPE-5 is still in COCO format, while YOLO only supports its own normalized text format.

2. Class Mapping

```
# -- SH17 (17 kelas -> 12 kelas) --
ID_MAPPING = {
    0: 0,      # person      -> person
    1: None,   # ear         -> DIABAIKAN
    2: None,   # ear-mufs    -> DIABAIKAN
    3: 1,      # face       -> face
    4: None,   # face-guard -> DIABAIKAN
    5: 2,      # face-mask  -> face-mask
    6: 3,      # foot       -> foot
    7: None,   # tool       -> DIABAIKAN
    8: 4,      # glasses    -> glasses
    9: 5,      # gloves     -> gloves
    10: 6,     # helmet     -> helmet
    11: 7,     # hands      -> hands
    12: 8,     # head       -> head
    13: 9,     # medical-suit -> body-protection
    14: 10,    # shoes      -> shoes
    15: 9,     # safety-suit -> body-protection
    16: 11,    # safety-vest -> safety-vest
}
```

SH17 serves as the main dataset (baseline) containing 17 classes.

We selected only 12 classes relevant to Personal Protective Equipment (PPE), removed irrelevant classes (ear, ear-muffs, face-guard, and tool), and merged medical-suit and safety-suit into a single class called body-protection.

The remaining datasets were then mapped into the same 12 classes to ensure consistency across all datasets.



DATA PRE-PROCESSING

3. Data Merging Using Symbolic Links

```
def process_and_copy(img_path, lbl_path, mapping, split_name, prefix, copy_labels=True):  
    """Membuat symlink gambar dan menulis ulang label dengan pemetaan kelas."""  
    new_img_name = f"{prefix}_{img_path.name}"  
    sym_img = WORK_DIR / f"images/{split_name}" / new_img_name  
  
    if not sym_img.exists():  
        os.symlink(img_path, sym_img)  
        split_lines[split_name].append(str(sym_img))  
  
    if copy_labels and lbl_path.exists():  
        dest_lbl = WORK_DIR / f"labels/{split_name}" / f"{prefix}_{lbl_path.name}"  
        new_annotations = []
```

All images from the **8 datasets** were consolidated into a single centralized directory without duplicating physical files, reducing storage usage and simplifying the training process.

4. 60% Labeled / 40% Unlabeled Data Split

```
random.seed(42)  
train_list = list(train_stems)  
random.shuffle(train_list)  
n_labeled = int(len(train_list) * 0.60)  
labeled_stems = set(train_list[:n_labeled])  
unlabeled_stems = set(train_list[n_labeled:])
```

The **semi-supervised** learning strategy requires a portion of unlabeled data for pseudo-labeling. Therefore, the SH17 dataset was randomly split using a fixed seed to ensure reproducibility.



FEATURE ENGINEERING

1. Minor Class Oversampling

```
MINOR_OVERSAMPLE = {
    2: 2,    # face-mask
    3: 2,    # foot
    9: 4,    # body-protection (prioritas utama)
    5: 2,    # gloves
    10: 2,   # shoes
    4: 2,    # glasses
}

# Gandakan gambar dan label
for dup_idx in range(1, max_factor):
    new_stem = f"{lbl_path.stem}_os{dup_idx}"
    new_img = train_image_dir / f"{new_stem}{img_path.suffix}"
    new_lbl = train_label_dir / f"{new_stem}.txt"

    if not new_img.exists():
        src = os.readlink(img_path) if img_path.is_symlink() else img_path
        os.symlink(src, new_img)
    if not new_lbl.exists():
        new_lbl.write_text(lbl_path.read_text())
```

Balancing the class distribution to prevent the model from being biased toward majority classes. Rare classes such as face-mask and body-protection were oversampled.

2. Augmentation Strategy (Two Phases)

```
# Phase 1 – Supervised
copy_paste = 0.8
mixup      = 0.25
close_mosaic = 10
```

Since only **60% of the data is labeled**, aggressive augmentation was applied to prevent overfitting and train invariant features.

```
# Phase 2 – SSL (Freeze Backbone)
freeze      = 10
copy_paste = 0.3
mixup      = 0.10
close_mosaic = 5
```

- **Lightweight augmentation and backbone freezing** were applied to prevent the model from learning noise from pseudo-labels.
- Mosaic augmentation was disabled during the last **5 epochs**.



FEATURE ENGINEERING

3. Adaptive Threshold Pseudo-Label Filtering

```
# — Threshold per-kelas (dikalibrasi dari mAP Phase 1) —  
PER_CLASS_CONF = {  
  0: 0.60, # person      (mAP 0.896) — ketat  
  1: 0.58, # face        (mAP 0.900) — ketat  
  2: 0.30, # face-mask   (mAP 0.778) — longgar  
  3: 0.40, # foot        (mAP 0.888) — moderat  
  4: 0.35, # glasses     (mAP 0.736) — longgar  
  5: 0.30, # gloves      (mAP 0.681) — longgar  
  6: 0.65, # helmet      (mAP 0.977) — sangat ketat  
  7: 0.52, # hands       (mAP 0.861) — moderat  
  8: 0.58, # head        (mAP 0.898) — ketat  
  9: 0.35, # body-protection (mAP 0.820) — longgar  
 10: 0.28, # shoes       (mAP 0.651) — paling longgar  
 11: 0.52, # safety-vest (mAP 0.882) — moderat  
}
```

Class-wise adaptive thresholds (**KETAT** / **MODERAT** / **LONGGAR**) were applied to reduce noise in pseudo-labels. Easy classes (e.g., helmet, mAP 0.977) were assigned high confidence thresholds, while difficult classes (e.g., shoes, mAP 0.651) were given lower thresholds to ensure pseudo-label generation without introducing excessive false positives.



GROUP 4

MODEL DEVELOPMENT



MODEL DEVELOPMENT FLOW

The main model uses YOLOv9-c with a two-phase training strategy.

Phase 1 - Supervised

Training YOLOv9-c on 60%
labeled data.
25 epochs • batch size 32 • image
size 640 • SGD optimizer • dual
Tesla T4 GPUs

Pseudo-labeling

The Phase 1 model predicts unlabeled
data. Predictions are accepted as
labels if the confidence exceeds the
class-specific threshold.

Phase 2 - SSL

Retraining is performed using a
combination of original labeled
data and pseudo-labels to
improve generalization across
PPE classes.

**The SSL strategy enables the model to learn from unlabeled data
without increasing manual annotation costs.**



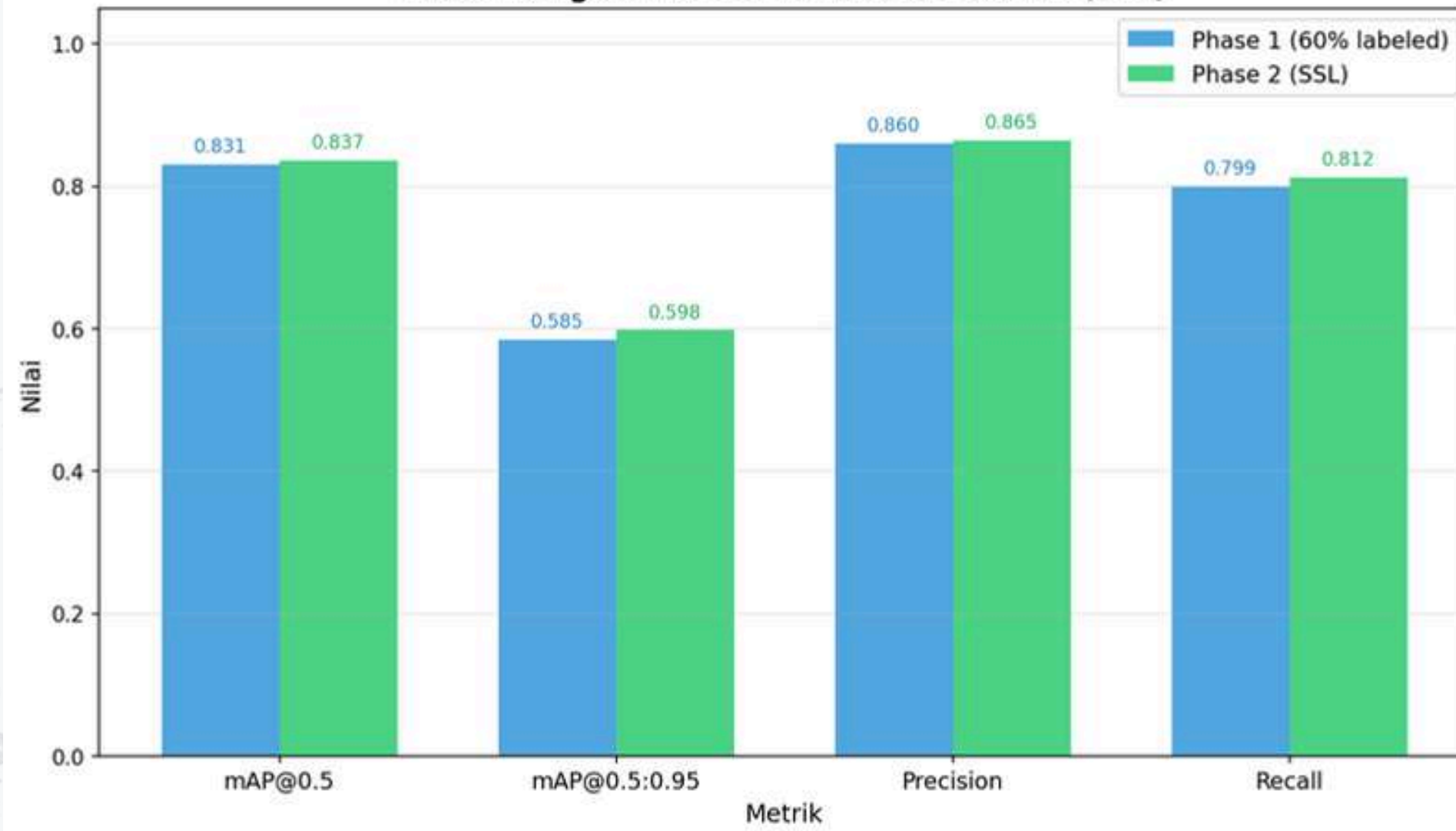
GROUP 4

MODEL EVALUATION



MODEL EVALUATION

Perbandingan Metrik Phase 1 vs Phase 2 (SSL)



Overall mAP@0.5: 0.837

- **mAP@0.5:0.95 improvement (+0.0132):** shows that the Phase 2 model achieves better localization accuracy across multiple IoU thresholds, not only at IoU 0.5.
- **Precision improvement (+0.0053):** suggests that pseudo-labels generated by the teacher model are of good quality and do not introduce significant false positives.
- **Recall improvement (+0.0127):** indicates that the SSL model is better at detecting PPE objects that were previously missed.



MODEL EVALUATION

2. PER-CLASS mAP@0.5 – PHASE 1 vs PHASE 2

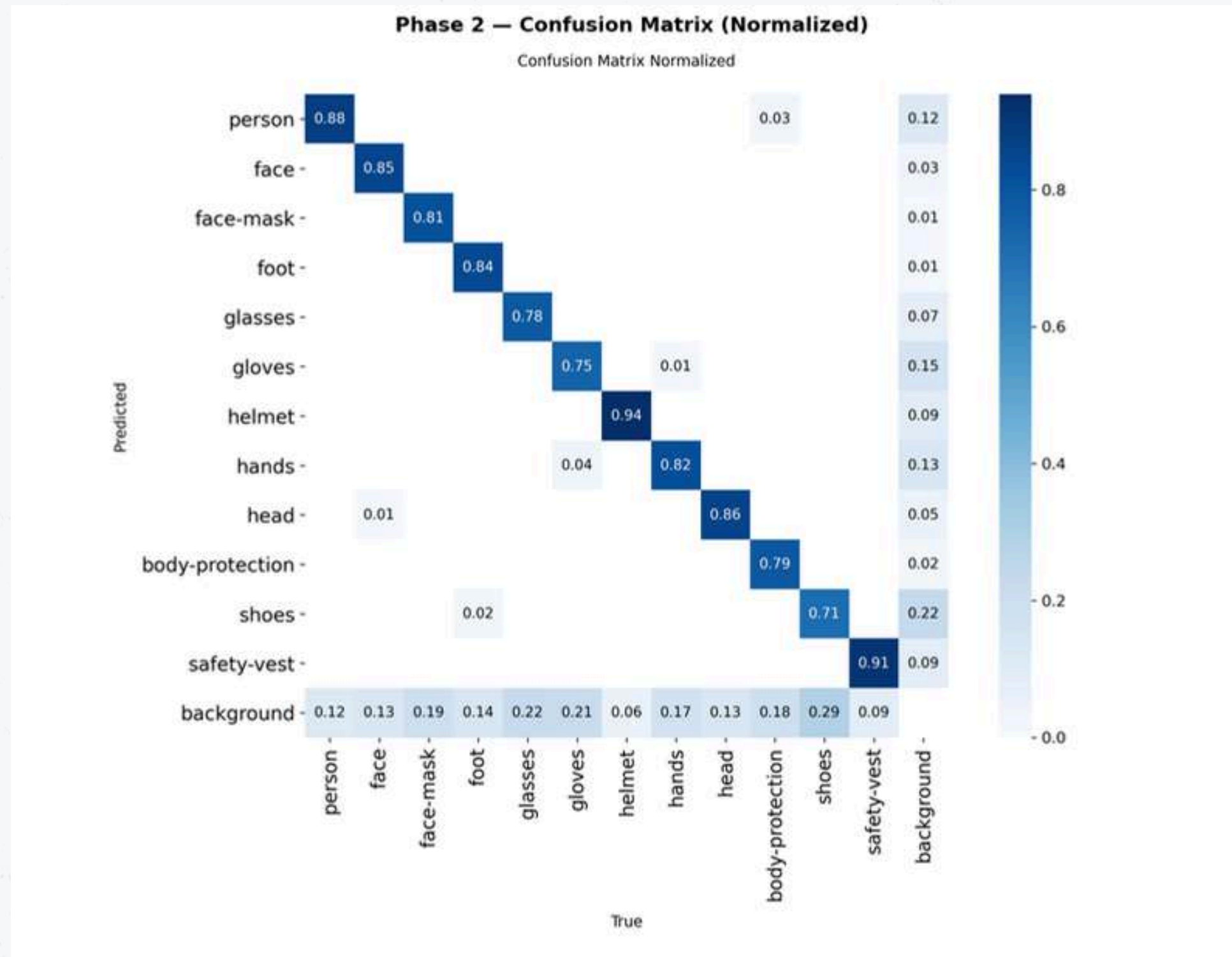
Class	Phase 1	Phase 2	Delta	Trend
person	0.896	0.901 +	0.005	↑ IMPROVED
face	0.900	0.902 +	0.002	↑ IMPROVED
face-mask	0.778	0.801 +	0.023	↑ IMPROVED
foot	0.888	0.895 +	0.007	↑ IMPROVED
glasses	0.736	0.746 +	0.010	↑ IMPROVED
gloves	0.681	0.688 +	0.007	↑ IMPROVED
helmet	0.977	0.974 -	0.003	↓ degraded
hands	0.861	0.860 -	0.001	↓ degraded
head	0.898	0.901 +	0.003	↑ IMPROVED
body-protection	0.820	0.858 +	0.038	↑ IMPROVED
shoes	0.651	0.650 -	0.001	↓ degraded
safety-vest	0.882	0.868 -	0.014	↓ degraded

8 out of 12 classes showed an improvement.



GROUP 4

MODEL EVALUATION

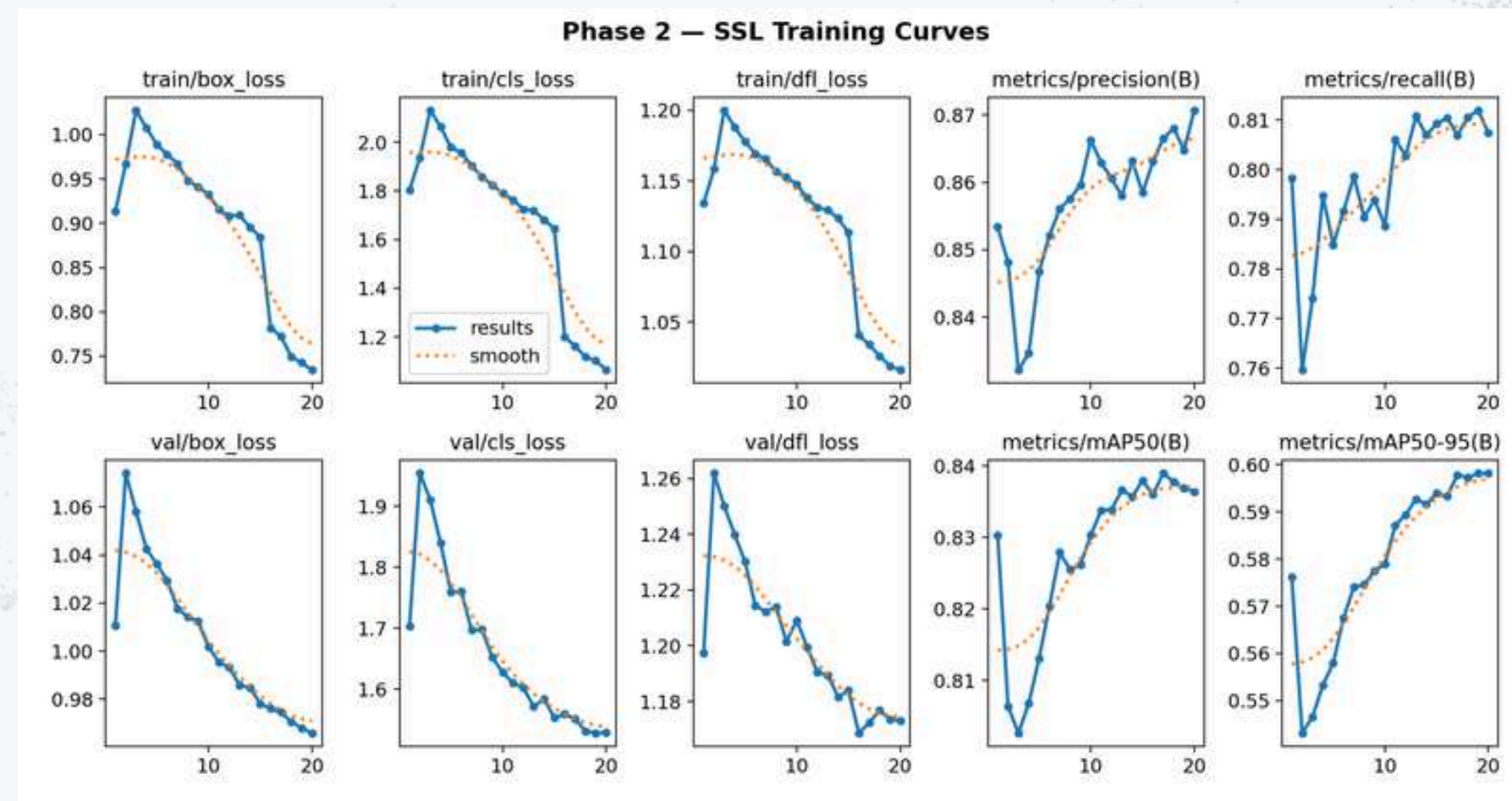


The confusion matrix shows a strong dominance of the main diagonal (> 0.8) **for 8 out of 12 classes**, indicating that the model can effectively distinguish PPE classes.



GROUP 4

MODEL EVALUATION

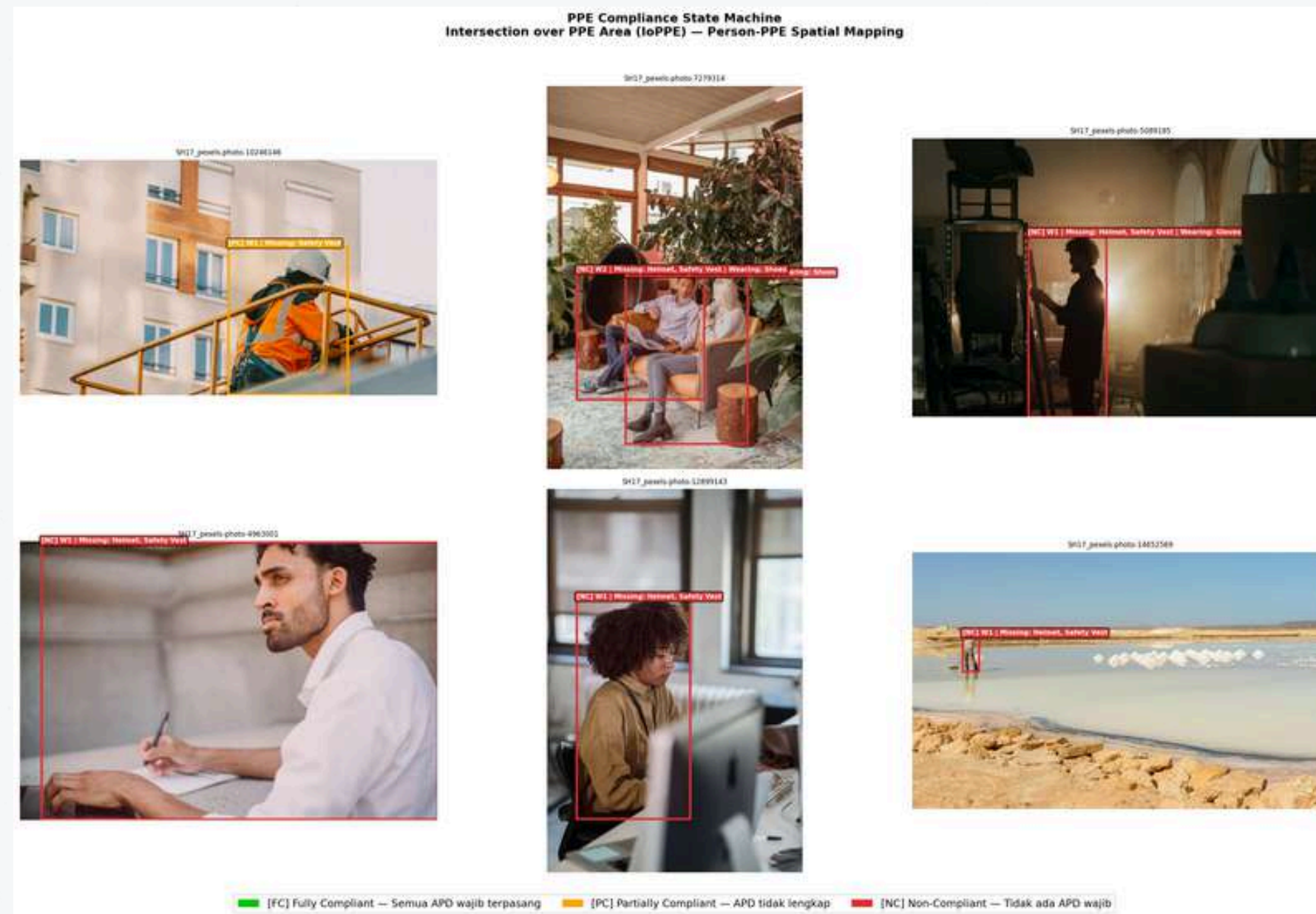


Training curves show stable convergence with no signs of overfitting, and the model achieves its best performance at epoch 17.



GROUP 4

COMPLIANCE STATE MACHINE



Proposed a Compliance State Machine using IoPPE-based spatial mapping to evaluate PPE compliance at the individual-worker level, enabling robust classification into **Fully Compliant**, **Partially Compliant**, and **Non-Compliant** states.



GROUP 4

MODEL VALIDATION



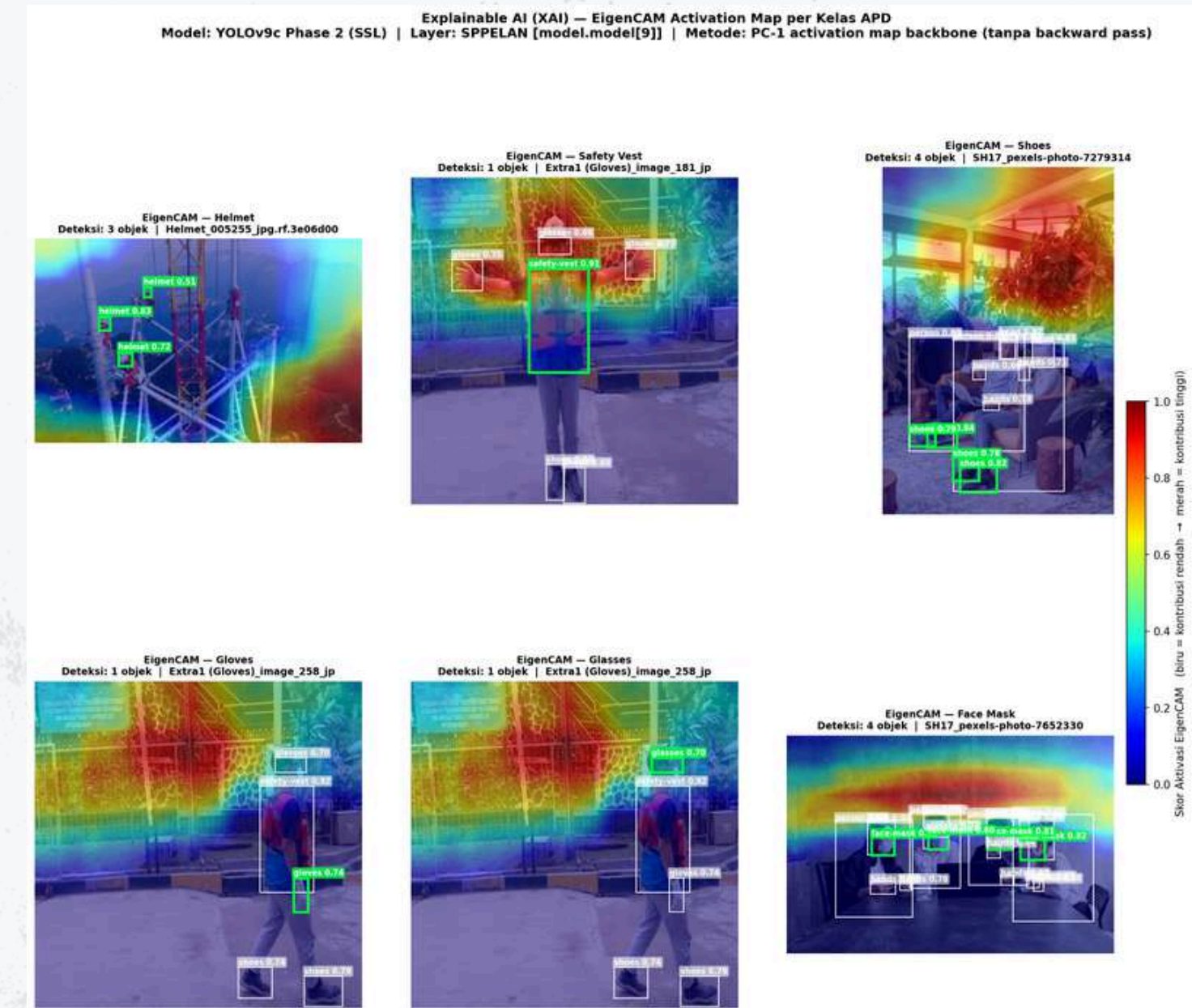
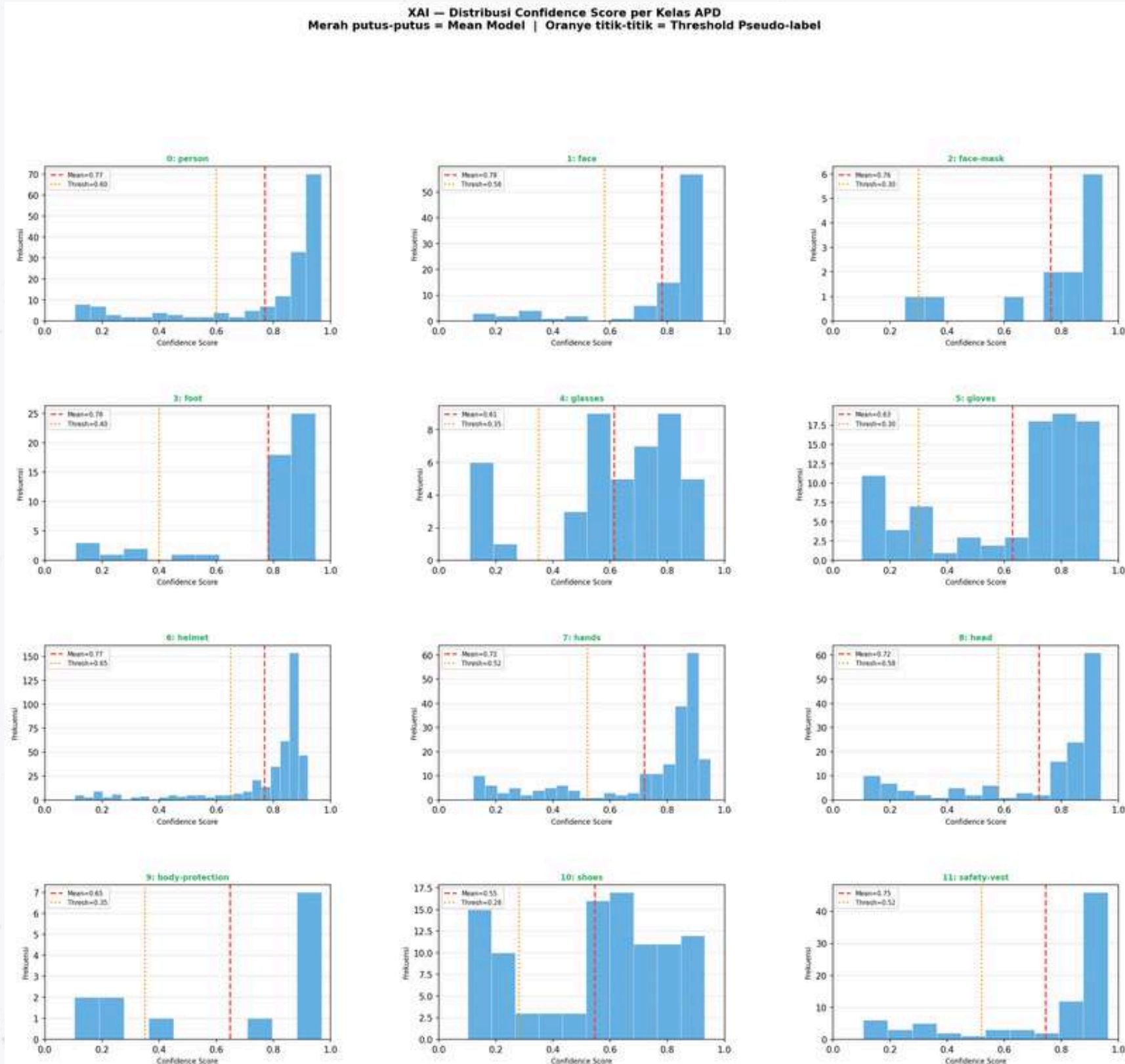
GROUP 4

MODEL VALIDATION

using Explainable AI

EigenCAM Visualization

EigenCAM was selected because it is gradient-free and more stable for object detection compared to Grad-CAM, which requires a single class score.



XAI & FUTURE WORK

Transparency

EigenCAM and confidence analysis make model decisions easier to audit.

Limitations

Small object classes such as shoes and gloves are more vulnerable to occlusion, blur, and extreme poses.

Future Development

Add active learning, CCTV video data, and real-time evaluation for deployment in construction environments.

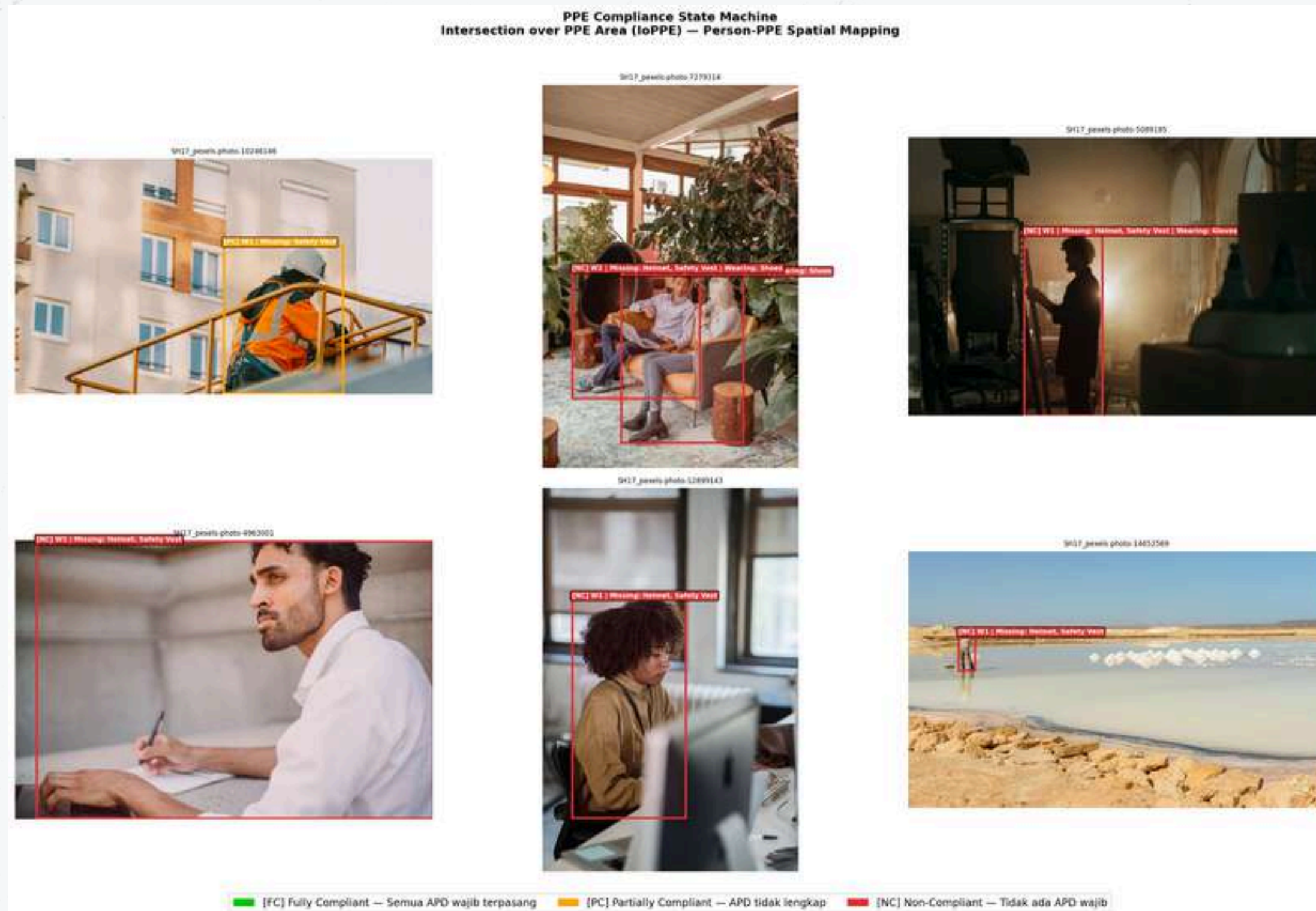
COMPETITIVE GAP

Main Focus	Gap
Only covers one to five types of PPE and does not comprehensively evaluate compliance levels across various PPE types.	Comprehensive 12 PPE Classes: Classifies objects ranging from large to small (Safety Helmet, High-Visibility Vest, Gloves, Boots, Glasses, Hearing Protection, etc.).
Fully dependent on fully supervised learning; requires large annotated datasets; lacks labeling efficiency.	Semi-Supervised Learning (SSL): Enables efficient model training using only 60% labeled data.
Only detects PPE without a systematic or rule-based mechanism for analyzing worker compliance status.	Compliance State Machine: A logical reasoning system used to determine per-worker compliance status.
Does not implement Explainable AI (XAI); operates as a black-box model with no component for explaining AI decision-making.	Explainable AI (XAI): Internal model auditing using Eigen-CAM visualization.



GROUP 4

MODEL RESULTS





GROUP 4

THANK YOU
SO MUCH!